



Novel compact device for clinically measuring extraocular muscle (EOM) tension



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ARTICLE INFO

Article history:

Accepted 8 July 2020

Keywords:

Extraocular muscle
Forced duction test
Strabismus
Tension measurement

ABSTRACT

Accurate knowledge of extraocular muscle (EOM) tension is important for the diagnosis of and surgical planning for strabismus, such as choosing which eye to operate or determining the amount of muscle displacement. Previous evaluations of passive EOM tension have relied extensively on the experience and skill of ophthalmic surgeons, who generally perform such evaluations by gripping the eyeball and then pushing and pulling it. This methodology, named the forced duction test, has the significant limitation that the tension is felt subjectively via the forceps, with the results therefore not being quantifiable. Previous quantitative analyses have utilized several different types of equipment with implanted force transducers or have involved connecting the muscle tendon to a strain gauge. However, the associated equipment setups and recording systems are highly complex and rarely used outside research settings. This situation prompted the present study to develop a novel compact, quantifiable and clinically applicable device for measuring the passive tension in human EOMs for use in clinical practice. The device employs locking forceps and a tilting sensor to rule out effects of the gripping force and to compensate for changes in the force due to tilting, which improves the measurement accuracy. The performance of the device was investigated in 60 eyes of 30 consecutive anaesthetized patients immediately prior to ophthalmic surgery. The results showed that the measured EOM tension in each rectus muscles agreed with previous findings: 48.3 ± 14.5 g (0.82 ± 0.28 g/deg, mean $\pm$ SD) for the lateral rectus, 45.6 ± 13.2 g (0.82 ± 0.23 g/deg) for the medial rectus, 48.6 ± 14.7 g (0.71 ± 0.21 g/deg) for the inferior rectus and 53.4 ± 13.7 g (0.77 ± 0.25 g/deg) for the superior rectus.

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1. Introduction

Strabismus is a condition in which the visual axes of the eyes are not parallel, resulting in the eyes appearing to be looking in different directions. If strabismus is left untreated, double vision, abnormal head posture, decreased vision or eye strain can develop (Franzblau, 1977). Strabismus can be corrected by adjusting the extraocular muscle (EOM) tension so as to align the visual axes, which can be achieved by either strengthening (by muscle resection) or loosening (by reattaching the muscle at a new location) the appropriate eye muscles (Kushner and Vrabec, 1987). Previous studies have found that the amount of muscle displacement

required for surgery or choosing which eye to operate should be determined with consideration of the tension in the EOMs (Vishwanath and Ansons, 2009; Kim et al., 2018). Thus, evaluating the passive tension in EOMs is important for the diagnosis of and surgical planning for strabismus (Jung and Holmes, 2015).

Evaluations of the passive EOM tension are currently highly dependent on the experience and skill of ophthalmic surgeons, who generally perform such evaluations by gripping and moving the eyeball with the aid of forceps, all the while maintaining the closing force on the forceps in order to maintain the grip. This methodology, named the forced duction test, has the significant limitation that the tension is felt subjectively via the forceps while moving the eyeball, which could therefore easily differ between surgeons even in the same individual (Metz, 1976; Alan et al., 1972). This situation results in inaccuracy and variability in measurements of the passive EOM tension, and hence also can lead

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to inadequate strabismus surgery (Kim et al., 2018). Reducing such errors requires precise and objective measurements of the EOM tension.

Previous quantitative analyses of EOM tension have utilized several different types of equipment with implanted force transducers (Collins et al., 1975) or have involved connecting the muscle tendon to a strain gauge (Collins et al., 1981; Lennerstrand et al., 2006). However, the associated equipment setups and recording systems are highly complex and rarely used outside research settings. In addition, the obtained data have largely come from strabismic eyes, which might not be representative of normal subjects. There is an obvious need for data to be obtained from a comprehensive sample of normal individuals.

We have designed a novel device for measuring the passive tension in human EOMs that can be easily deployed in clinical practice. The proposed EOM tension-measuring device (ETMD) could provide normative measurements of passive tension in EOMs based on the moment transfer of force sensors. The obtained values provide a quantitative baseline of the passive tension in EOMs in the normal eye and may facilitate the ability of clinicians to make differential diagnoses, thereby resulting in more-accurate quantitative planning of strabismus surgery.

2. Results and discussion

2.1. Design of the ETMD

We have developed a novel compact ETMD (Fig. 1) that comprises three units: (i) sensing unit, (ii) control unit and (iii) display unit. A miniaturized compressive-type load cell (EX-601D-10, Caskorea, Gyeonggi-do, Korea), a MEMS-based biaxial tilt sensor (SA2, DAS, Gyeonggi-do, Korea) and Castroviejo locking forceps (Storz E-1798-S, Karl Storz, Tuttlingen, Germany) were employed for the sensing unit. The operating range for the load cell and tilt sensor were from 0 to 5 kg and from -60 to 60 deg, respectively. In the main body of the sensing unit (Fig. 1(a)), the tailored support block assembling the Castroviejo locking forceps is attached to the hinge structure. Then, an external force acting on the tip of the forceps generates a rotating moment around the hinge. The other end of the support block is placed above the compressive-type load cell to measure the transferred EOM tension, resulting in the rotating moment. A standard double-shielded ball bearing (682ZZ, NSK, Japan) was employed for the frictionless rotating motion in the hinge.

The control unit comprises a DAQ (data acquisition) board (USB-6001, National Instruments, TX, USA), laptop (Desktop 400

G2, Hewlett-Packard, CA, USA) and amplifier (AL-50, Caskorea) for data collection and storage. The amplified analogue EOM tension signal as well as the two tilt angles of the sensing unit are converted and stored in the control unit in digital form using the commercially available LabVIEW software (National Instruments). The sampling frequency of the ADC (analogue-to-digital converter) was 20 Hz in our study, with a 14-bit resolution. The sensitivity threshold of the measuring unit for tension was 0.305 g and for the tilt angle was 0.0037 deg. Also, a first-order low-pass filter was applied to remove high-frequency noise. The display unit includes a USB-powered portable LCD monitor (AOC1659, Alphascan, Seoul, Korea) to display the EOM tension signal as well as the tilt angles of the sensing unit in both the moving and perpendicular directions. The calculated EOM tension and tilt angle are displayed in a user interface, as shown in Fig. 2(a).

2.2. EOM tension-measurement procedure

A standardized anesthetic regimen was applied to all patients by a single anesthesiologist to produce a consistent depth of anesthesia. All patients arrived at the operation room without anticholinergic premedication. Anesthesia was induced by administering 5 mg/kg thiopental sodium and 0.6 mg/kg rocuronium (muscle relaxant). After performing tracheal intubation, anesthesia was maintained by administering sevoflurane with 40% oxygen in air. The anesthetic depth was monitored by using bispectral index in all patients and the values of it were maintained for range from 40 to 60 during study period.

Before measuring the EOM tension, the load cell and amplifier were calibrated to ensure precise sensing using counterweights, as shown in Fig. 2(b). The correlation between the load cell output and counterweight was calculated as

$$y = 26.592x - 1.1224 \quad (1)$$

where y is the mass in grams and x is the load-cell output in volts. R^2 was 99.94% for Eq. (1), which showed that the calibration was sufficient precise for EOM tension measurements.

The signal flow of the ETMD is shown in Fig. 2(c). The EOM tension measured by the load cell is amplified and collected by the DAQ board, whereas the tilt angles are directly connected to the DAQ board (i.e. bypassing the amplifier). Collected data on the tension and tilt angles are saved in text format as well as being displayed on graphs along with numerical values on the portable monitor in real time. This setup allows the surgeon to measure the EOM tension by carefully moving the sensing unit while maintaining an accurate posture.

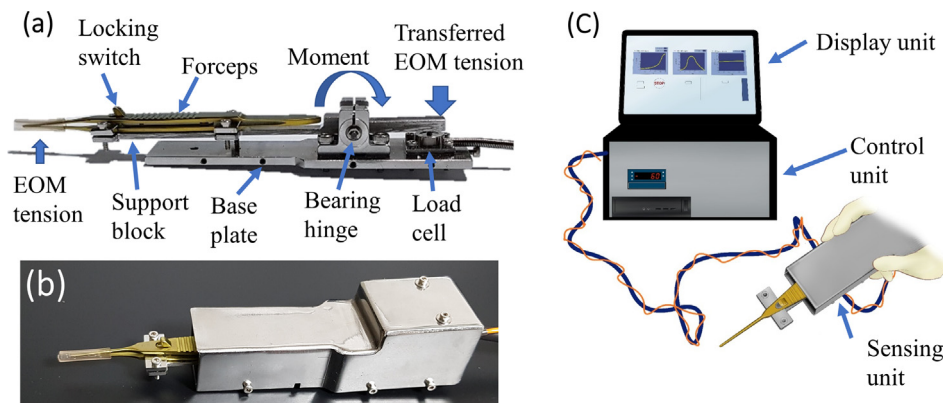


Fig. 1. Novel compact extraocular muscle (EOM) tension-measuring device (ETMD). (a) Detail of the sensing unit, comprising locking forceps, hinge, bearing, support block, base plate and load cell. Tension in the EOM is transferred as a force to the load cell by the hinge and bearing. (b) Appearance of the sensing unit with its cover on. (c) The ETMD comprised three units: (i) sensing unit, (ii) control unit and (iii) display unit.

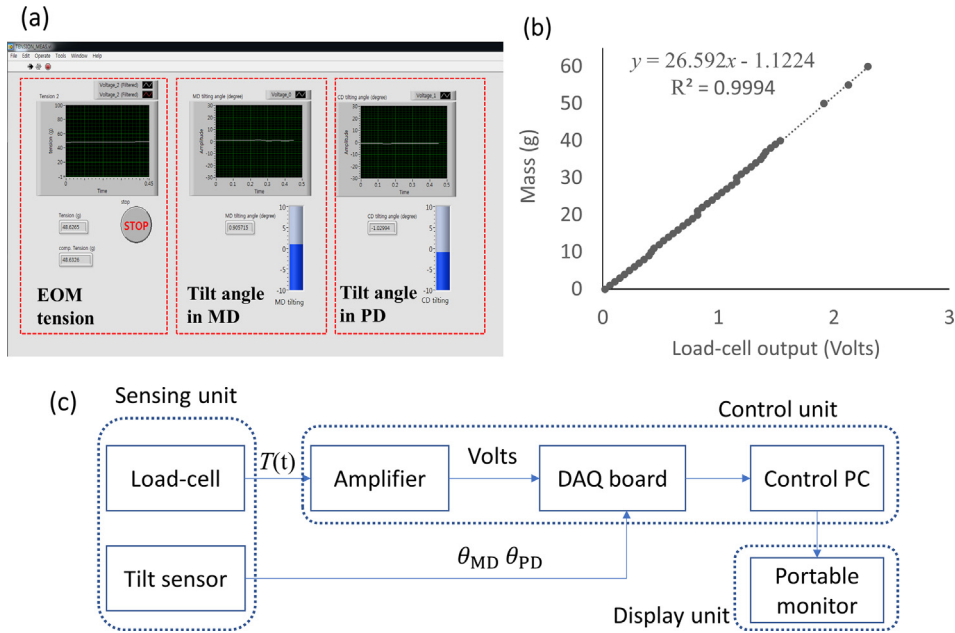


Fig. 2. (a) The user interface of the ETMD consists of three parts: (i) display of the EOM tension, (ii) display of the tilt angle of the sensing unit in the moving direction (MD) and (iii) display of the tilt angle of the sensing unit in the perpendicular direction (PD). (b) Calibration results between the output of the load cell and the counterweight. (c) Signal flow of the ETMD from the sensing unit to the display unit.

Four sets of measurements were made on each eye. Successively, Castroviejo locking forceps were attached to the eyeball just posterior to the limbus perpendicularly at the 3-, 6-, 9- and 12-o'clock positions. The eye was then rotated from the rest position under anaesthesia 12.5 mm (60 deg) away from the direction of force to be tested (Rosenbaum and Myer, 1980). A counterforce was then generated at the other end of the support block above the load cell. For example, the passive tension in the medial rectus (MR) was measured with the forceps attached to the eyeball just posterior to the medial limbus. The sensing unit of the ETMD was rotated in a lateral direction by the examiner (Fig. 3(b) and Supplementary Video 1). The measured tilt angle of the sensing unit was used to compensate for the angle at which the EOM tension was being applied to the load cell (Fig. 3(a)). The perpendicular direction force on the load cell was calculated from the applied EOM tension using the following equation:

$$T = T_p / \cos(\theta) \quad (2)$$

where T is the applied EOM tension, T_p is the force measured in the perpendicular direction and θ is the tilt angle of the sensing unit. Fig. 3(a) illustrates the EOM tension applied to the forceps and the sensing unit.

In addition, based on the assumptions that (i) the eyeball is a perfect sphere, (ii) the rotating centre of the eyeball is fixed and (iii) the mean ocular diameter is 25 mm, we calculated the degree of eyeball rotation from the eyeball moving distance as (Chesnut, 2018; Lim et al., 2014)

$$\theta_r = 2\sin^{-1}(d/D) \quad (3)$$

where θ_r is the angle of eyeball rotation, d is the eyeball moving distance and D is the diameter of the eyeball.

The proposed ETMD was used to measure the passive tension in EOMs in the 60 eyes of 30 consecutive anaesthetized patients immediately prior to ophthalmic surgery. This clinical study was approved by the Institutional Review Board/Ethics Committee at Konkuk University Medical Center (registration number: KUH1100071). The study was conducted according to the principles expressed in the Declaration of Helsinki, including obtaining

written informed consents from all of the individual participants included in the study.

All of the patients were within 2 prism diopter of being orthophoric without any limitations of eye movement, and confirmed as American Society of Anesthesiologists physical status classification I. Exclusion criteria were the presence of other ocular diseases (e.g. strabismus, nystagmus, ptosis, blepharospasm or glaucoma), thyroid disorder, muscular or neurological diseases (e.g. Parkinson's disease or myasthenia gravis), history of ocular trauma, previous strabismus or ocular surgery, or a history of receiving medications known to affect muscle tension (e.g. muscle relaxants).

2.3. Statistical analysis

All calculations and statistical analyses were performed using Minitab (version 19.2, Minitab, State College, PA, USA). The Anderson-Darling test was used to determine whether the data conformed to a parametric (Gaussian) or non-parametric (non-Gaussian) distribution. ANOVA was used to perform comparisons among the four rectus muscles. The data are presented as mean $\pm$ SD values, and the criterion for statistical significance was set as $p < 0.05$.

The subjects were aged 4 to 80 years, with a mean age of 31.8 years, and comprised 9 males and 21 females. Eye movements were induced by the examiner applying external forces, and so the tensions were generated in a passive state. The ETMD continuously measured and recorded the EOM tensions during the eye movements, which manifested as S-shaped curves (Fig. 3(c)). The maximum tension in each rectus was recorded. The tension data for all of the EOMs conformed to a normal distribution according to the Anderson-Darling test.

The passive tension across the four rectus muscles was 49.0 ± 14.2 g (0.73 ± 0.30 g/deg). Fig. 3(d) shows the passive tension of each rectus muscle, which was 48.3 ± 14.5 g (0.82 ± 0.28 g/deg) for the lateral rectus (LR), 45.6 ± 13.2 g (0.82 ± 0.23 g/deg) for the MR, 48.6 ± 14.7 g (0.71 ± 0.21 g/deg) for the inferior rectus (IR) and 53.4 ± 13.7 g (0.77 ± 0.25 g/deg) for the superior rectus (SR). The passive

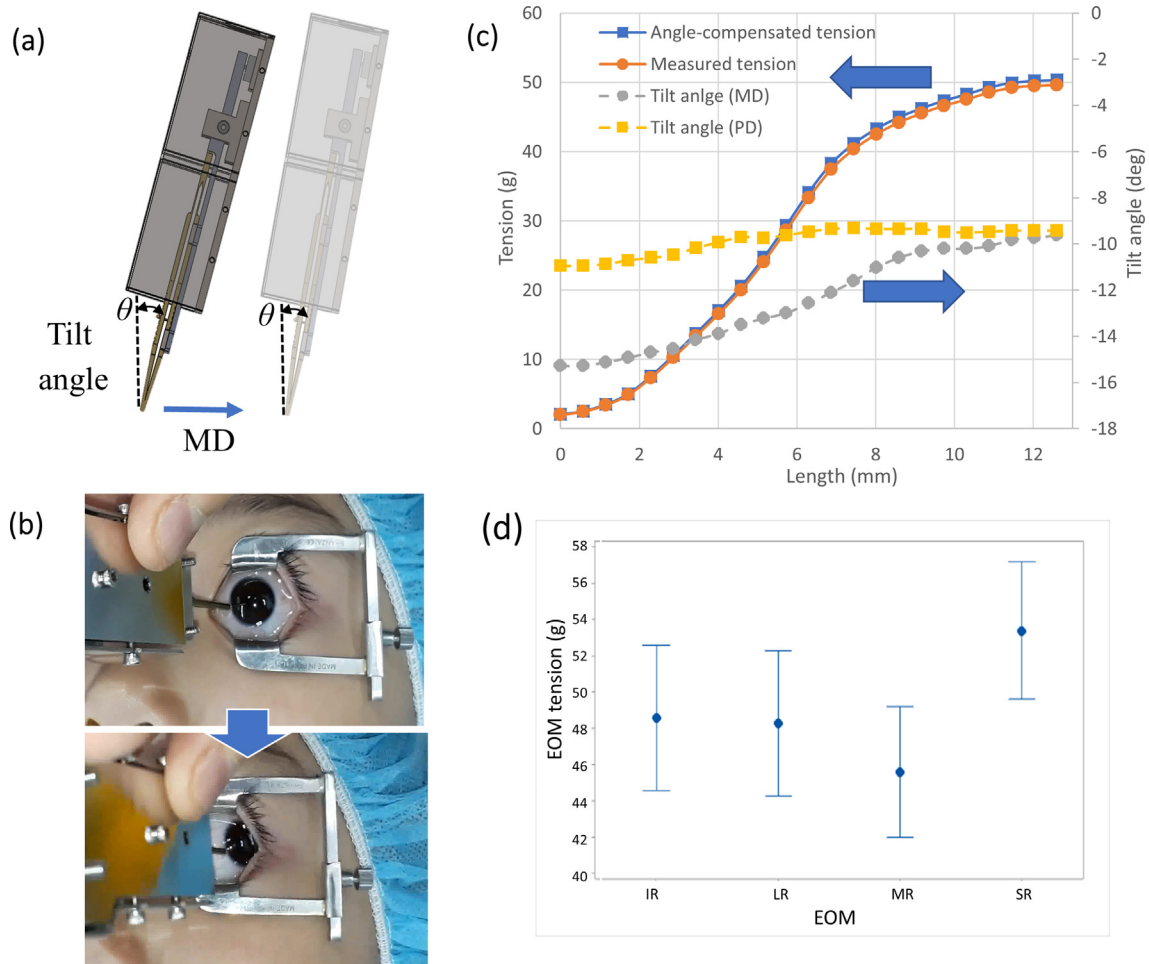


Fig. 3. (a) Schematic of passive tension measurements of EOMs with the tilt angle in the MD. (b) For measuring the medial rectus (MR) of the left eye, the 9-o'clock limbus was held by the ETMD forceps, and then moved to the lateral side. (c) Representative length–tension and tilt-angle records for the lateral rectus (LR) of the right eye, demonstrating an S-shaped length–tension curve. The other rectus muscles showed similar patterns. (d) Summary data of the EOM tensions for the MR, LR, inferior rectus (IR) and superior rectus (SR). Data are mean and SD values.

tensions were similar in the LR, MR and IR, while it was larger in the SR than in the MR ($P = 0.003$). Our results are similar to previous reports of the passive EOM tensions measured using different methods (Collins, 1971; Collins et al., 1975; Scott, 1971; Gao et al., 2014). The mean passive tension across the four rectus muscles has previously reportedly ranged from 40 to 50 g, and our mean value of 49 g falls within this range. In addition, the mean passive tensions of the LR, MR, IR and SR have previously been reported to be similar to each other (Rosenbaum and Myer, 1980; Stephens and Reinecke, 1967), which is also consistent with the present findings.

3. Conclusion

In conclusion, we have developed a novel device to facilitate measurements of the passive tensions in human EOMs in clinical practice. The proposed device is compact, quantifiable and clinically applicable. The utilization of locking forceps and a tilting sensor avoids confounding effects of the gripping force and compensates for changes in the force due to tilting, thereby improving the accuracy of the measurements. The performance of the new device was investigated in 60 eyes of 30 consecutive anaesthetized patients immediately prior to ophthalmic surgery, which yielded measured values of the tension in all rectus muscles that were consistent with previous findings. We believe that the

proposed ETMD can be used to obtain normative measurements of the EOM tension, as well as be deployed in clinical practice.

4. Contributorship Statement

Hyun Jin Shin: the conception of the study, acquisition of data, and drafting the article, **Sang Jae Lee:** analysis and interpretation of data, **Chung-Sik Oh:** revising the article, **Hyunkyoo Kang:** the design of the methodology and software, drafting the article, and final approval of the version to be submitted.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by Basic Science Research Program through the National Research Foundation of Korea, South Korea (NRF) funded by the Ministry of Education (2020R111A3075301).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbiomech.2020.109955>.

References

- Alan, B.S., Carter, C.C., David, M.O., 1972. A Forceps to measure strabismus forces. *Arch. Ophthalmol.* 88, 330–333.
- Collins, C.C., 1971. Orbital mechanics. In: Bach-Y-Rita, P., Collins, C.C., Hyde, J.E. (Eds.), *The Control of Eye Movements*. Academic Press, New York, pp. 283–325.
- Collins, C.C., O'Meara, D., Scott, A.B., 1975. Muscle tension during unrestrained human eye movements. *J. Physiol.* 245, 351–369.
- Collins, C.C., Carlson, M.R., Scott, A.B., Jampolsky, A., 1981. Extraocular muscle forces in normal human subjects. *Invest. Ophthalmol. Vis. Sci.* 20, 652–664.
- Chesnut, C., 2018. Restoration of visual loss with retrobulbar hyaluronidase injection after hyaluronic acid filler. *Dermatol. Surg.* 44, 435–437.
- Franzblau, S.C., 1977. The diagnosis of amblyopia and strabismus: a programmed approach. *J. Am. Optom. Assoc.* 48, 1248–1257.
- Gao, Z., Guo, H., Chen, W.C.C., 2014. Initial tension of the human extraocular muscles in the primary eye position. *J. Theor. Biol.* 21 (353), 78–83.
- Jung, J.H., Holmes, J.M., 2015. Quantitative intraoperative torsional forced duction test. *Ophthalmology* 122, 1932–1938.
- Kim, S., Yang, H.K., Hwang, J.M., 2018. Surgical outcomes of unilateral recession and resection in intermittent exotropia according to forced duction test results. *PLoS ONE* 13, e0200741.
- Kushner, B.J., Vrabec, M., 1987. Theoretical effects of surgery on length tension relationships in extraocular muscles. *J. Pediatr. Ophthalmol. Strabismus* 24, 126–131.
- Lennerstrand, G., Schiavi, C., Tian, S., Benassi, M., Campos, E.C., 2006. Isometric force measured in human horizontal eye muscles attached to or detached from the globe. *Graefes Arch. Clin. Exp. Ophthalmol.* 244, 539–544.
- Lim, H.W., Lee, D.E., Lee, J.W., Kang, M.H., Seong, M., Cho, H.Y., Oh, J.E., Oh, S.Y., 2014. Clinical measurement of the angle of ocular movements in the nine cardinal positions of gaze. *Ophthalmology* 121, 870–876.
- Metz, H.S., 1976. Forced duction, active force generation, and saccadic velocity tests. *Int. Ophthalmol. Clin.* 16, 47–73.
- Rosenbaum, A.L., Myer, J.H., 1980. New instrument for the quantitative determination of passive forced traction. *Ophthalmology* 87, 158–163.
- Scott, A.B., 1971. Extraocular muscle forces in strabismus. In: Bach-Y-Rita, P., Collins, C.C., Hyde, J.E. (Eds.), *The Control of Eye Movements*. Academic Press, New York, pp. 327–342.
- Stephens, K.F., Reinecke, R.D., 1967. Quantitative forced duction. *Trans. Am. Acad. Ophthalmol. Otolaryngol.* 71, 324–329.
- Vishwanath, M., Ansons, A., 2009. Midoperative forced duction test as a guide to titrate the surgery for consecutive exotropia. *Strabismus* 17, 41–44.